Applications of First-Order Linear Equations

MATH 146 – Calculus II

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Exponential Growth

Example: The rate at which a viral outbreak is spreading is proportional to the number of people who are already infected. On the first day, 10 people are infected and five days later, 100 people are infected. Find a formula for how many people will be infected on day t. When will 10,000 people be infected?

Mixing Problems

Example: Initially 5 grams of salt are dissolved in 20 liters of water. Brine with concentration of salt 2 grams of salt per liter is added at a rate of 3 liters per minute. The tank is mixed well and is drained at 3 liters a minute. How long does the process have to continue until there are 20 grams of salt in the tank?

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Newton's Law of Cooling

Let $T(t)$ be the temperature of an object at time t and let A be the (constant) ambient/room temperature. Then the rate at which the temperature $T(t)$ is changing is proportional to the difference between the ambient temperature and the the temperature T.

Example: A fluid with a temperature of $100°C$ is placed outside on a day when the temperature is $-10°C$. After one minute the temperature of the fluid is $80°C$. Find a formula for the temperature of the fluid at time t.